

Comprehensive Disease Characterization

Report: Melorheostosis

Disease: Melorheostosis (Léri disease / Léri-Joanny syndrome) **MONDO ID:** MONDO:0007970

Category: Somatic mosaic sclerosing bone dysplasia with segmental hyperostosis **Report date:**

2026-09-08 **Evidence base:** ~35 primary papers and reviews (PMIDs cited inline). Evidence source types are marked: *[human clinical]*, *[in vitro]*, *[model organism]*, *[review]*.

Summary (Answer to the Research Question)

Melorheostosis is an ultra-rare (~0.9 per million), sporadic, non-hereditary sclerosing bone dysplasia characterized by asymmetric, segmental (sclerotomal) cortical and medullary hyperostosis that produces the pathognomonic radiographic "dripping candle wax" appearance, typically in the appendicular skeleton, accompanied by pain, deformity, joint contractures, and overlying soft-tissue changes. It is caused predominantly by **post-zygotic somatic mosaic gain-of-function mutations in MAP2K1 (MEK1)** that hyperactivate the RAS-MAPK/ERK pathway in osteoblast-lineage cells, with genetically distinct subtypes driven by somatic *SMAD3* (TGF- β /SMAD), *KRAS*, and — in the osteopoikilosis/Buschke–Ollendorff spectrum only — germline *LEMD3*. There is no cure; management is symptom-directed (NSAIDs, bisphosphonates, physiotherapy, surgery), with MEK- and CDK4-inhibitor targeted therapy emerging from preclinical work.

1. Disease Information

Overview. Melorheostosis is a rare, benign, progressive sclerosing bone dysplasia of mesodermal origin featuring exuberant cortical bone overgrowth, most often in a single limb, that flows along the bone like melted wax. It was first described by Léri and Joanny (1922). *[review]* "Melorheostosis is a rare bone disease characterized by abundant bone formation with a characteristic radiographic appearance that resembles 'dripping candle wax'" (

Key identifiers. - **MONDO:** MONDO:0007970 - **OMIM:** 155950 (Melorheostosis, isolated) - **Orphanet:** ORPHA:2485 - **ICD-10:** Q78.8 (Other specified osteochondrodysplasias); **ICD-11:** LD24.Y / FB80.Y (other specified osteopathies) — often coded under other specified osteochondrodysplasias - **MeSH:** D008586 (Melorheostosis; MeSH tree C05.116.099.708.410) - **UMLS/SNOMED CT:** Melorheostosis (disorder), SCTID 43994001

Synonyms / alternative names. Léri disease; Léri–Joanny syndrome; "flowing hyperostosis"; "candle wax bone disease"; rheostosis; osteosis eburnisans monomelica.

Source of information. Derived predominantly from **aggregated disease-level resources** (OMIM, Orphanet) and from **individual-patient case reports/case series** plus a single institution's prospective **natural-history cohort** (NIH; PMIDs 31485554, 35403375). There is no large EHR/registry dataset given the rarity.

2. Etiology

Primary causal factor: genetic (somatic mosaicism). Melorheostosis is caused by **post-zygotic (somatic) mosaic activating mutations** arising in mesenchymal/osteoblast-lineage precursors during development. There is **no infectious, environmental, or dietary cause**; it is not inherited.

Genetic risk factors (causal variants). - **MAP2K1 (MEK1)** — the major driver. Recurrent activating missense variants cluster in the MEK1 negative-regulatory domain: **p.Gln56Pro (Q56P)**, **p.Lys57Glu (K57E)**, **p.Lys57Asn (K57N)**, and a catalytic-domain variant **p.Cys121Ser (C121S)** (PMIDs 29643386, 32387835). [*human clinical / in vitro*] "we identify somatic mosaic MAP2K1 mutations in affected, but not unaffected, bone of eight unrelated patients ... The activating mutations (Q56P, K57E and K57N) cluster tightly in the MEK1 negative regulatory domain"

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P 29643386
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- **SMAD3** — somatic activating mutations in **endosteal-pattern** melorheostosis

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- **KRAS** — somatic activating variants in a subset (osteopathia-striata-like pattern)

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P 30989250
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- **LEMD3 (MAN1)** — **germline** loss-of-function mutations cause osteopoikilosis (OPK) and Buschke–Ollendorff syndrome (BOS); they do **not** cause isolated sporadic melorheostosis (PMIDs 17087626, 19438932, 31129707).

Modifier genes. None firmly established. Because lesions are clonal and mosaic, disease extent likely reflects the developmental timing and location of the somatic mutation rather than trans-acting modifiers.

Environmental risk factors. None identified (toxins, occupation, radiation, age, sex, family history all non-contributory). Trauma is sometimes an incidental discovery trigger, not a cause (

P 41938408

Protective factors. None identified (genetic or environmental).

Gene–environment interactions. None documented; the disease is monogenic-somatic.

3. Phenotypes

Melorheostosis is highly variable ("variable severity, progressive/insidious course"). Core phenotypes with suggested HPO terms and frequency (from the 47-patient natural-history cohort and case series; PMIDs 31485554, 35403375, 39776616):

Phenotype	Type	HPO term	Onset	Severity/ Course	Frequency
Bone/limb pain	Symptom	Bone pain HP:0002653	Childhood– adult	Chronic, progressive	Very frequent (presenting complaint)
Cortical hyperostosis / sclerosis	Physical sign (imaging)	Cortical thickening of long bones HP:0005791; Hyperostosis HP:0100774	Congenital lesion, symptoms later	Progressive	Defining feature (~100%)
Joint contracture	Clinical sign	Joint contracture HP:0034392	Childhood– adult	Progressive	Frequent
Limited range of motion	Clinical sign	Limitation of joint mobility HP:0001376	Variable	Progressive	Frequent
Limb deformity / length discrepancy	Physical	Limb deformity; Limb undergrowth	Childhood	Progressive	Frequent
Limb swelling / soft- tissue mass	Sign	Localized skin lesion; Soft tissue swelling	Variable	Stable/ progressive	Common
Overlying skin changes (hyperpigmentation, sclerotic/nevoid skin, hemangioma)	Physical	Abnormality of the skin HP:0000951	Variable	Stable	Subset
Neurological compromise (nerve entrapment,	Sign	Peripheral neuropathy; Myelopathy	Variable	Progressive	Rare (axial disease)

Phenotype	Type	HPO term	Onset	Severity/ Course	Frequency
myelopathy in spinal disease)					
Fatigue (general/physical)	Symptom	Fatigue HP:0012378	Adult	Chronic	Prominent functional burden

Laboratory abnormalities: characteristically **absent** — serum calcium, phosphate, alkaline phosphatase, ESR, CRP are **normal** (PMIDs 42559563, 41938408). Normal biochemistry is itself a diagnostic clue.

Age of onset. Lesions are congenital/developmental but frequently present in childhood or adulthood; late-onset monomelic presentations occur (P 42559563).

Quality-of-life impact. [human clinical] Substantial functional burden: high-demand leisure activities were the least retained and most often given up (27%); general and physical fatigue were the most limiting constructs, with physical fatigue moderately negatively correlated with activity engagement ($r = -0.524$, $p < .001$) (P 35403375).

4. Genetic / Molecular Information

Causal genes (gene symbol; HGNC; OMIM; locus). - **MAP2K1** — HGNC:6840; NCBI Gene 5604; OMIM 176872; 15q22.31; encodes MEK1 (UniProt Q02750). Major driver. - **SMAD3** — HGNC:6769; NCBI Gene 4088; OMIM 603109; 15q22.33. Endosteal subtype. - **KRAS** — HGNC:6407; NCBI Gene 3845; OMIM 190070; 12p12.1. Rare subset. - **LEMD3** — HGNC:28887; NCBI Gene 23592; OMIM 607844; 12q14.3. Germline; OPK/BOS spectrum (not isolated MEL).

Pathogenic variants. - **Type/class:** missense, gain-of-function (activating). **MAP2K1:** Q56P, K57E, K57N (negative-regulatory domain, exon 2 hotspot); C121S (catalytic domain). - **Classification:** Pathogenic (activating) per functional evidence; these are established oncogenic hotspots (also seen in Langerhans-cell histiocytosis, arteriovenous malformations,

and some cancers). - **Somatic vs germline: Somatic/mosaic**, present in affected bone and overlying skin but **not** in unaffected tissue or blood (

P 29643386

mosaicism detected in overlying skin in 4/5 patients tested. Skin distribution of somatic variants was mapped in

P 32791068

- **Allele frequency:** Absent from population databases (gnomAD/1000G) as constitutional variants — they are somatic and, as germline events, would be embryonic-lethal or oncogenic.
 - **Functional consequence:** Gain of function → constitutive MEK1 kinase activity → increased phospho-ERK1/2 (*MAP2K1*); enhanced TGF- β /SMAD transcriptional activity (*SMAD3*).

Modifier genes / epigenetics / chromosomal abnormalities. No established modifier genes, no recurrent epigenetic signature, and **no chromosomal abnormalities** (aneuploidy/translocation/CNV) are implicated. A subset of clinically classical cases has **no identifiable mutation** in *MAP2K1/SMAD3/LEMD3/KRAS*

P 32387835

indicating additional undiscovered drivers.

5. Environmental Information

- **Environmental factors:** None implicated (no toxin, radiation, pollution, or occupational association).
- **Lifestyle factors:** None (no smoking/diet/alcohol/exercise association).
- **Infectious agents: Not applicable** — melorheostosis is not infectious.

This section is largely not applicable: melorheostosis is a somatic genetic disease with no known environmental contribution.

6. Mechanism / Pathophysiology

Ordered causal chain

MAPK branch (classical "dripping candle wax", MAP2K1): 1. A **post-zygotic somatic MAP2K1 mutation** arises in an osteoblast/mesenchymal precursor during development → **leads to** a mosaic clone of mutant cells confined to a sclerotome/limb segment. 2. The mutation (e.g., K57N) **causes** constitutive MEK1 kinase activity (loss of negative autoregulation). 3. Constitutive MEK1 **results in** increased phospho-ERK1/2 signaling, detectable as a mosaic pattern (two osteoblast populations with distinct p-ERK levels) (

[P 29643386](#)

4. ERK hyperactivation **leads to** increased osteoprogenitor proliferation via cell-cycle activation (elevated phospho-Rb, Ki-67, higher S-phase fraction; CDK4-dependent) (PMIDs 29643386, 41395834). 5. Mutant cells **secrete increased VEGF**, driving hypervascularity/angiogenesis; VEGF is *sufficient alone* to increase mineralization of osteogenic cultures (

[P 37737377](#)

[*in vitro*] 6. In parallel, the mutation **impairs BMP2-mediated terminal mineralization/differentiation**, which **results in** accumulation of excess unmineralized osteoid alongside dense cortical/woven bone (

[P 29643386](#)

7. The net effect **results in** segmental cortical/medullary hyperostosis → clinically pain, deformity, contractures, limb swelling; paradoxically the bone is *harder and mechanically stronger* despite decreased mineral density (inferred from geometry/microstructure; (

[P 31689526](#)

[*model organism*]

TGF-β/SMAD branch (endosteal pattern, SMAD3): 1b. A **somatic SMAD3-activating mutation** → **enhances** TGF-β/SMAD nuclear translocation and target-gene expression in osteoblasts (

[P 32232430](#)

2b. This **stimulates** osteoblast differentiation and mineralization (antagonized by BMP2) → **endosteal-pattern** hyperostosis.

(*KRAS-driven cases converge on RAS–MAPK activation, upstream of MEK1.*)

Detail by category

- **Molecular pathways:** RAS–MAPK/ERK cascade (KEGG hsa04010; Reactome R-HSA-5673001 "RAF/MAP kinase cascade"); TGF- β /SMAD signaling (KEGG hsa04350; Reactome R-HSA-170834). BMP signaling is antagonistic/dysregulated. GO: MAPK cascade GO:0000165; ERK1/ERK2 cascade GO:0070371; transforming growth factor beta receptor signaling pathway GO:0007179.
- **Cellular processes:** increased cell proliferation (GO:0008284), dysregulated osteoblast differentiation (GO:0001649) and ossification (GO:0001503), angiogenesis (GO:0001525). Cell-cycle dysregulation (phospho-Rb/CDK4). No apoptosis or inflammation signature is prominent.
- **Protein dysfunction: Gain of function** — constitutively active MEK1 kinase; hyperactive SMAD3 transcription factor. Not misfolding/aggregation.
- **Metabolic changes:** No systemic metabolic derangement (normal Ca/PO₄/ALP). Local: increased collagen I/IV secretion by mutant iMSCs
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- **Immune system involvement:** Not a primary feature; co-expression clustering noted altered interferon signaling in SMAD3 lesions
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— significance uncertain.
- **Tissue damage mechanisms:** Mechanical (mass effect → contracture, nerve/cord compression), not oxidative/ischemic/fibronectrotic.
- **Molecular profiling:** RNAseq/GSEA of lesional tissue shows **upregulation of proliferative pathways**
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and TGF- β /ossification/ECM-organization programs
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Proteomic: elevated VEGF and collagen secretion
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- **Functional genomics:** iPSC→iMSC→osteoblast patient models and MEK1DD mouse (see §15).

Cell types (CL): osteoblast CL:0000062; osteoprogenitor/mesenchymal stem cell CL:0000134; fibroblast CL:0000057 (overlying skin carries the mutation). **Chemical entities (CHEBI):** zoledronic acid CHEBI:46557; palbociclib CHEBI:85993; trametinib CHEBI:75998 (MEK inhibitor class).

7. Anatomical Structures Affected

- **Organ level (primary):** Bone (UBERON:0002481 bone tissue; UBERON:0001474 bone element), predominantly **long bones of the limbs** (appendicular skeleton). Lower-extremity long bones (femur, tibia, fibula) most common; upper extremity (humerus, radius, ulna, hand) frequent. Cohort: 30/47 lower extremity, 14/47 upper extremity, 1 both (P 35403375).
- **Secondary involvement / body systems: Musculoskeletal** (joints — contracture; muscle/soft tissue — swelling, sclerosis, ossification), **integumentary** (overlying skin — hyperpigmentation, scleroderma-like/nevoid changes), **nervous** (peripheral nerve entrapment; spinal cord compression/myelopathy in rare axial disease, (P 41852826), **vascular** (hemangioma/hypervascularity). Axial sites reported: spine (C-spine, (P 41852826), pubis (P 42047352).
- **Tissue/cell level:** cortical and endosteal **bone** (dense cortical, woven bone, hypervascular); connective tissue of skin (collagen/elastin). Target cells: osteoblasts (CL:0000062), osteoprogenitors/MSCs (CL:0000134).
- **Subcellular (GO cellular component):** nucleus GO:0005634 (SMAD3 translocation), cytoplasm/plasma membrane (MEK1/ERK signaling GO:0005886), extracellular region/matrix GO:0005576 (osteoid, collagen).
- **Localization/UBERON:** femur UBERON:0000981; tibia UBERON:0000979; fibula UBERON:0001446; humerus UBERON:0000976; radius UBERON:0001423; ulna UBERON:0001424; vertebral column UBERON:0001130.

- **Lateralization:** Characteristically **unilateral, asymmetric, monomelic**, in a **sclerotomal** distribution; bilateral/multifocal is uncommon (PMIDs 41938408, 39776616).

8. Temporal Development

- **Onset:** Congenital/developmental lesion (somatic mutation occurs in utero); clinical onset is typically **insidious**, presenting in childhood or adulthood; late-onset adult presentations occur.
- **Onset pattern:** Chronic, insidious (not acute).
- **Progression:** **Slowly progressive**, lifelong. Bone overgrowth and contractures accrue over years to decades (e.g., 20-year progressive history, [P 42555496](#)).
- No formal staging system exists.
- **Course pattern:** Progressive/stable; not relapsing-remitting or episodic.
- **Duration:** Chronic, lifelong (not self-limited).
- **Remission:** No spontaneous remission of bone lesions; symptom control (pain) is achievable with treatment.
- **Critical periods:** Developmental window (timing/location of the somatic mutation) determines lesion distribution/extent; there is no defined therapeutic "critical window," though early physiotherapy may limit contractures.

9. Inheritance and Population

- **Epidemiology — prevalence:** ultra-rare, **~0.9 per million** (i.e., ~0.00009 per 100,000); global prevalence < **1 per million** (PMIDs 41938408, 42559563). Incidence not reliably estimated. *[review/case series]* "an estimated prevalence of 0.9 per million" ([P 41938408](#)).
- **Inheritance pattern:** **Sporadic, non-hereditary** — driven by somatic mosaicism; **not** Mendelian. (Germline *LEMD3* → autosomal dominant OPK/BOS is a separate, related entity.)

- **Penetrance/expressivity:** Not applicable in the classical sense (somatic); lesion expressivity is highly variable.
- **Genetic anticipation / germline mosaicism / founder effects / consanguinity / carrier frequency:** **Not applicable** — no vertical transmission, no carriers, no founder or consanguinity effect.
- **Population demographics:** No ethnic or geographic predilection; reported worldwide. **Sex ratio ~1:1** (no strong sex predilection). **Age distribution:** diagnosed across pediatric to older-adult ages.

10. Diagnostics

Diagnosis is primarily radiographic, supported by normal biochemistry and (non-specific) histology; molecular confirmation requires **lesional tissue**.

- **Imaging (cornerstone):** Plain radiograph/CT — flowing cortical **hyperostosis resembling "dripping/melting candle wax"** in a sclerotomal distribution (pathognomonic). *[review/case]* "a pathognomonic imaging appearance of flowing hyperostosis resembling melted candle wax"
([P 32994853](#)).
- **MRI** delineates soft-tissue involvement/extent; **Tc-99m-MDP bone scintigraphy** shows increased uptake reflecting activity
([P 42555496](#));
- **FDG/PSMA PET** may be positive incidentally (PMIDs 42313078, 42047352). RadLex: hyperostosis.
- **Laboratory tests/biomarkers:** No diagnostic blood/urine biomarker; serum Ca, PO₄, ALP, ESR, CRP **normal** (LOINC panels for these analytes). Normality helps exclude Paget disease and metabolic bone disease.
- **Biopsy/histopathology (supportive, non-specific):** dense cortical bone (73.3%), woven bone (60%), hypervascularity/increased porosity (66.7%), prominent cement lines (33%)
([P 31386640](#));
hyalinized collagen with dystrophic calcification also described

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Biopsy is often *avoidable* when imaging is classic.

- **Genetic testing: Sequencing of affected lesional tissue (bone and/or overlying skin), not blood**, because variants are somatic/mosaic. WES, WGS, and RNAseq of affected-vs- unaffected tissue are the definitive research/clinical approach and identified *MAP2K1/ SMAD3*/novel variants (PMIDs 29643386, 32232430, 32387835). Targeted single-gene/panel testing (*MAP2K1*, *SMAD3*, *KRAS*, *LEMD3*) on lesional DNA is reasonable. Karyotype/CMA/ FISH/mtDNA/repeat testing: **not indicated** (no such abnormalities).
- **Omics-based diagnostics:** RNAseq of lesion (proliferative-pathway upregulation) is research-grade; no validated clinical omics/liquid-biopsy assay exists.
- **Clinical criteria / differential diagnosis:** No formal consensus criteria; diagnosis is clinicoradiologic. **Differential:** osteoma/osteoid osteoma, **osteoblastoma** (can mimic and recur —

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),

osteosarcoma/parosteal osteosarcoma, fibrous dysplasia, **Paget disease of bone**, osteopathia striata, **osteopoikilosis**, and mixed sclerosing bone dysplasia; distinguishing features are the flowing candle-wax pattern, sclerotomal distribution, normal labs, and benign non-aggressive behavior (PMIDs 32994853, 41214899).

- **Screening:** No population/newborn/carrier screening is applicable (somatic, non-heritable).

11. Outcome / Prognosis

- **Survival/mortality:** **Benign** disease; **normal life expectancy**; not directly life-limiting. No disease-specific mortality data (deaths are not attributable to melorheostosis itself). Malignant transformation is a *theoretical* concern but has **not been clinically reported**

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- **Morbidity/disability:** Chief morbidity is **chronic pain, deformity, contracture, and reduced mobility**, producing disability and reduced participation in high-demand activities; fatigue is prominent

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Rare severe morbidity: neurological deficit from nerve entrapment or spinal cord compression (PMIDs 41852826, 41717511).

- **Quality of life:** Impaired via pain/fatigue/loss of leisure activities (natural-history cohort,

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No disease-specific validated QoL instrument; generic tools used (Activity Card Sort, Multidimensional Fatigue Inventory, Lower/Upper Extremity Functional scales).

- **Complications:** contractures, limb-length discrepancy, joint dysfunction, soft-tissue ossification, fibrolipomatous/vascular masses, nerve/cord compression; rare fragility fracture at unaffected sites

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- **Recovery potential:** Bone lesions do not regress; symptoms are manageable. Function may improve with physiotherapy/analgesia and appropriately timed surgery.
- **Prognostic factors:** Lesion extent, axial (spinal) involvement (worse — neurological risk), degree of contracture, and pain severity. **Prognostic biomarkers:** none validated; *MAP2K1* vs *SMAD3* genotype correlates with radiographic subtype and may predict therapy response (predictive, not prognostic).

12. Treatment

No cure and no established guideline exist; management is multidisciplinary and symptom-directed

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[review/case] "There are no established guidelines for its management, as it is not a curable condition. Treatment primarily focuses on symptom relief."

- **Pharmacotherapy:**
- **NSAIDs** (e.g., celecoxib) for pain — VAS pain reduced to 1/10 with celecoxib + physiotherapy

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NCIT: Nonsteroidal Anti-inflammatory Agent.

- **Bisphosphonates** — IV **zoledronic acid** and **pamidronate** reduce pain and disease severity (PMIDs 42273437, 34169026, 32057643). NCIT: Bisphosphonate; Zoledronic Acid (CHEBI:46557).
- **Pharmacogenomics**: none specific.
- **Targeted / advanced therapeutics (experimental)**:
 - **MEK inhibitors** (trametinib/selumetinib class) — mechanistically rational (block MEK1→ERK); proposed by discovery authors as a treatment avenue
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NCIT: MEK Inhibitor.
 - **CDK4/6 inhibitor palbociclib** — FDA-approved; reduced phospho-Rb, proliferation, and mineralization in *MAP2K1*-mutant patient iMSCs/osteoblasts, "opening a pathway to treatment"
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[*in vitro*] NCIT: Cyclin-Dependent Kinase Inhibitor.
- Gene/cell/RNA therapy: none in use.
- **Surgical/interventional**: osteotomy/contracture release, excision of exostoses/soft-tissue masses, limb-lengthening/deformity correction, and **decompression/fusion** for spinal cord compression; reserved for severe deformity, refractory pain, or neurological compromise (PMIDs 41852826, 41938408). Recurrence after excision is common. NCIT: Orthopedic Surgery; Osteotomy.
- **Supportive/rehabilitative: Physical/occupational therapy** for range of motion, contracture prevention, and function; pain management; splinting (PMIDs 35403375, 41717511). NCIT: Physical Therapy; Rehabilitation Therapy.
- **Experimental / clinical trials**: An NIH natural-history study (NCT02504879) has characterized the disease; no approved disease-modifying drug trial has reported results. MEK/CDK4 inhibition remains preclinical.
- **Treatment outcomes/adverse events**: Bisphosphonate response is variable (only ~9 zoledronate reports to date;
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adverse events per drug class (bisphosphonate: acute-phase reaction, osteonecrosis of the jaw; MEK inhibitors: rash, ocular/cardiac toxicity; CDK4/6i: cytopenias).

- **Personalized medicine:** Genotype-guided targeting (MEK inhibitor for *MAP2K1*; potential TGF- β -directed approaches for *SMAD3*) is the emerging paradigm.

13. Prevention

- **Primary prevention: Not possible** — the causal somatic mutation arises sporadically in utero; no modifiable risk factor exists.
- **Secondary prevention (early detection):** Early radiographic recognition prevents misdiagnosis, unnecessary biopsy, and enables early physiotherapy to limit contractures (PMIDs 30647832, 34169026).
- **Tertiary prevention:** Physiotherapy, analgesia, bisphosphonates, and timely surgery to prevent/limit contracture, deformity, and neurological complications.
- **Immunization / behavioral / public-health / environmental interventions: Not applicable.**
- **Genetic counseling:** Reassurance that the disease is **sporadic and non-heritable**, with **negligible recurrence risk** to offspring or siblings (the mutation is somatic, not germline). Prenatal/carrier/PGD testing not applicable.

14. Other Species / Natural Disease

- **Taxonomy / natural disease:** Melorheostosis is essentially a **human-specific** clinical entity; a well-defined naturally occurring homolog in companion animals or wildlife is **not established** (no OMIA entry for a melorheostosis phenotype). Human: *Homo sapiens* NCBI:txid9606.
- **Orthologous genes:** *Map2k1* (mouse, NCBI Gene 26395; NCBI:txid10090), *smad3* (mouse 17127), *Kras* (mouse 16653) — highly conserved.
- **Comparative biology / evolutionary conservation:** The RAS–MAPK and TGF- β /SMAD pathways are deeply conserved across vertebrates; the disease mechanism is modeled in mouse (see §15). **Zoonotic potential:** none (not transmissible).

This section is largely not applicable as a naturally occurring cross-species disease; relevance is via engineered model organisms.

15. Model Organisms

- **Mammalian (mouse): MEK1DD conditional transgenic mouse** expressing constitutively active MEK1 in osteoprogenitors postnatally. *[model organism]* Recapitulates "extra-cortical bone formation, abundant osteoid formation, decreased mineral density, and increased porosity," with paradoxically stronger bone due to altered geometry/microstructure ([P 31689526](#)).
Model type: conditional/transgenic gain-of-function (Cre-driven in osteoprogenitors).
Resources: MGI (*Map2k1*, MGI:1346866).
- **Limitations:** does not reproduce the **mosaic/segmental (sclerotomal)** distribution, the overlying soft-tissue/skin involvement, or the natural somatic-mutation timing; expression is broader than the human clonal patch.
- **Human in vitro / iPSC models:** Patient-derived iPSCs reprogrammed (Sendai) from *MAP2K1*-mutant **skin fibroblasts**, differentiated to **induced MSCs → osteoblasts**; retain the mutation, show elevated MEK1/ERK activity, increased **VEGF**, proliferation, collagen I/IV, and mineralization (PMIDs 37737377, 41395834). Isogenic unaffected fibroblasts serve as controls. Cellosaurus/ATCC: patient-specific lines (not standardized).
- **Applications:** dissecting MAPK→proliferation/VEGF mechanisms, testing MEK and CDK4/6 inhibitors (palbociclib), and modeling the mineralization paradox.
- **Model databases:** MGI (mouse), Alliance of Genome Resources.

Supported vs Refuted Hypotheses

- Supported:** - Somatic mosaic activating *MAP2K1* mutations cause classical melorheostosis via MEK1→ERK hyperactivation ([P 29643386](#)).
- ✓ - Genetic heterogeneity: *SMAD3*/TGF- β drives the endosteal subtype; *KRAS* a rare subset (PMIDs 32232430, 30989250). ✓ - VEGF is a sufficient downstream driver of increased bone mineralization in mutant cells ([P 37737377](#)).

- ✓ - MEK1DD mouse recapitulates core skeletal features (P 31689526).
- ✓ - CDK4 inhibition (palbociclib) reduces mutant-cell proliferation/mineralization (P 41395834).
- ✓

Refuted / not supported: - Germline *LEMD3* mutation causes isolated sporadic melorheostosis — **refuted**; *LEMD3* underlies OPK/BOS but not isolated MEL (PMIDs 17087626, 19438932, 31129707). - Environmental/infectious/inherited causation — **not supported**; disease is sporadic-somatic.

Limitations and Future Directions

- Evidence rests on **small cohorts and case reports** (ultra-rare disease); a subset of classical cases has **no identified mutation**, implying additional drivers to discover.
- **No controlled therapeutic trials**; bisphosphonate and targeted-therapy evidence is anecdotal/preclinical. MEK- and CDK4/6-inhibitor efficacy/safety in patients is unproven.
- Genotype–phenotype correlations (subtype-specific therapy) and single-cell/spatial characterization of the mosaic lesion are priority future work.

Key ontology annotations (summary)

- **MONDO:** MONDO:0007970 · **HPO:** HP:0002653, HP:0100774, HP:0005791, HP:0034392, HP:0001376, HP:0012378
- **Genes (HGNC):** MAP2K1 (HGNC:6840), SMAD3 (HGNC:6769), KRAS (HGNC:6407), *LEMD3* (HGNC:28887)
- **GO:** GO:0000165, GO:0070371, GO:0007179, GO:0001503, GO:0001649, GO:0008284, GO:0001525
- **CL:** CL:0000062 (osteoblast), CL:0000134 (MSC), CL:0000057 (fibroblast)
- **UBERON:** UBERON:0001474 (bone element), UBERON:0000981 (femur), UBERON:0000979 (tibia), UBERON:0001446 (fibula)
- **CHEBI:** CHEBI:46557 (zoledronic acid), CHEBI:85993 (palbociclib), CHEBI:75998 (trametinib)

- **NCIT:** Bisphosphonate Therapy, NSAID, MEK Inhibitor, CDK Inhibitor, Orthopedic Surgery, Physical Therapy

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